

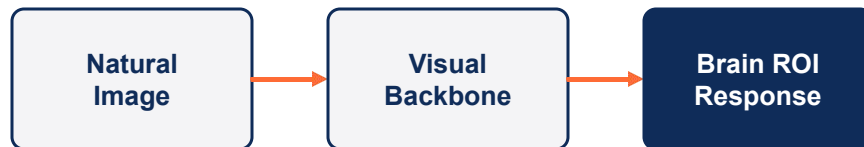
ROI- and Semantics-Aware Feature Pruning for Brain-Aligned Visual Encoding

Computer Vision Project

Students:

Antonio Serra 1893530

Riccardo Passacantando 1814248

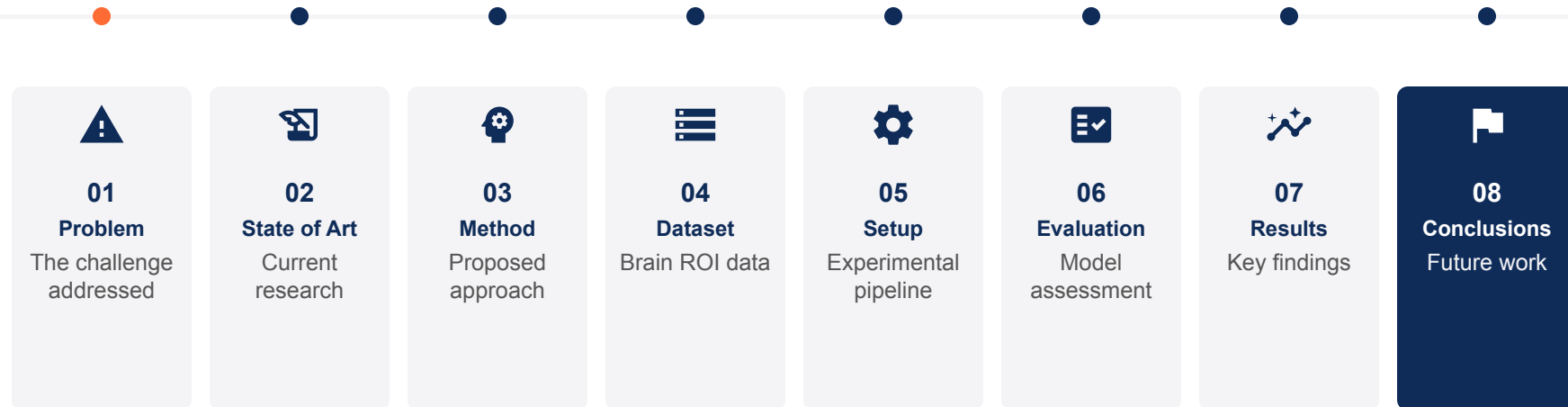


Academic Year: 2025/2026

Sapienza University of Rome

Presentation Outline

Computer Vision Project Workflow



Problem Statement

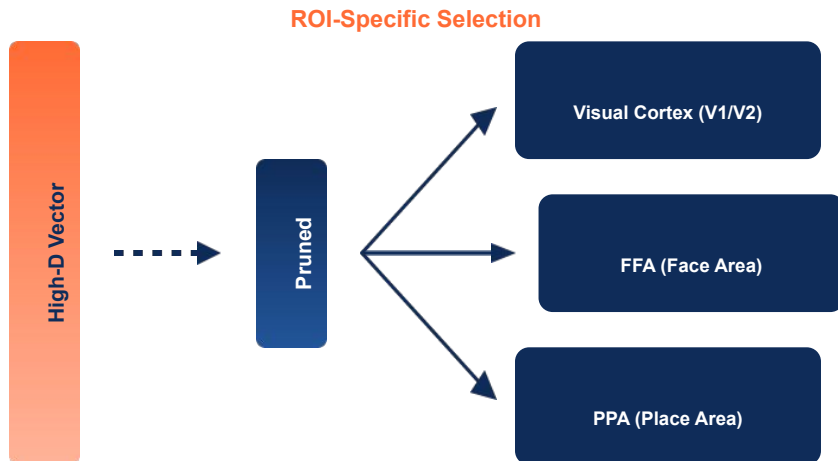
Modern visual backbones produce rich but high-dimensional feature representations.

- **Pretrained Visual Models:** DINOv2 and CLIP provide exceptionally strong pretrained visual representations.
- **Dimensionality Barrier:** These generated representations are extremely high-dimensional, making them difficult to interpret.
- **Functional Specialization:** Different cortical ROIs (Regions of Interest) may rely on distinct subsets of visual information.
- **Feature Pruning Concept:** We ask whether compact ROI-specific feature subsets can fully preserve fMRI brain prediction performance.



Key Question:

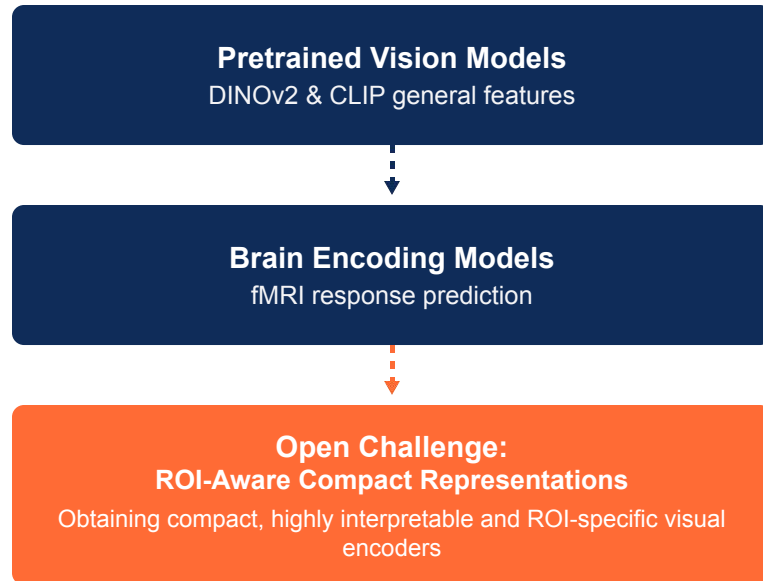
Can we reduce visual features while preserving ROI-level brain-encoding performance?



State of the Art and Research Inspiration

This project is inspired by recent work on brain-aligned visual encoders and compact neural representations.

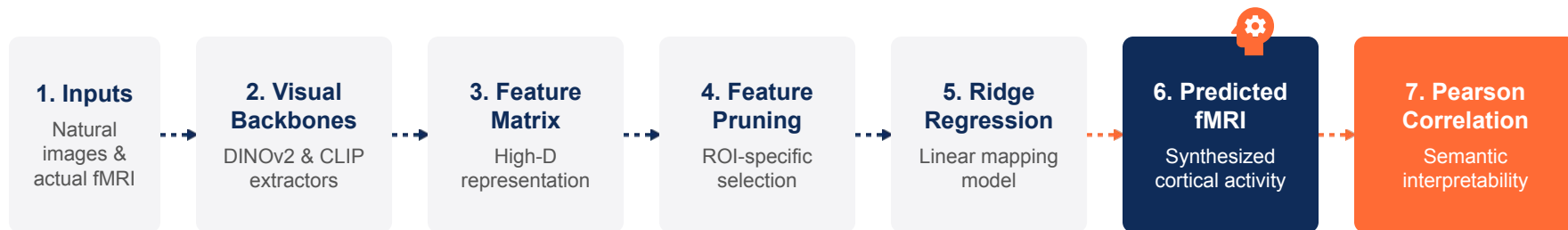
- **Brain Encoding Models:** Predict neural or fMRI responses using deep visual features.
- **Pretrained Encoders:** Models like DINOv2 and CLIP are increasingly utilized as feature extractors.
- **VISGate:** Explores ROI-conditioned visual encoders aligned with localized brain responses.
- **BOLDREAMS:** Investigates pruned, in-silico fMRI encoding models of the visual cortex.



Proposed Method: Full Pipeline

The notebook implements a complete visual encoding pipeline from raw inputs to evaluations.

- **Feature Extraction:** Extract frozen high-dimensional visual representations from DINOv2 and CLIP.
- **Pruning Strategies:** Select optimal feature subsets to reduce dimensionality and isolate key functional zones.
- **Ridge Regression:** Train linear encoding models to map features directly to fMRI brain responses.
- **Evaluation:** Benchmark predictions against actual ROI-level fMRI responses using Pearson correlation.



Dataset and fMRI Targets

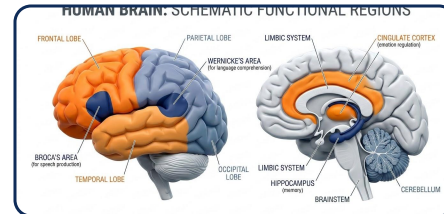
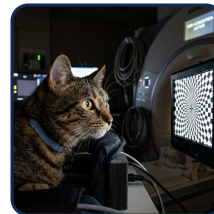
We model brain-aligned responses using curated visual cortex datasets mapped to semantic ROI structures.

Dataset Structure

- **Natural Images:** Sourced from NSD / Algonauts-style visual cortex datasets.
- **Human fMRI:** Clean, stimulus-associated BOLD cortical responses.

Target Definition

- **Scalar Output:** The model predicts exactly one scalar response per ROI.
- **ROI-Level Focus:** Bypasses unstable voxel prediction for robust, interpretable visual encoding.



Target Regions of Interest (ROIs)

Early Visual V1, V2, V3, hV4 Higher-Level OFA, FFA, OPA, PPA, RSC

Image
Natural Stimulus



ROI-level fMRI response
Target Brain Signal (Scalar)

Experimental Setup

Structured parameters and evaluation configurations for the brain-encoding pipeline.

Component	Configuration
Backbones	DINOv2 and CLIP, frozen feature extractors
Regressor	Ridge Regression
Training subject	Subject 01
Evaluation 1	Within-subject: train and test on Subject 01
Evaluation 2	Cross-subject: train on Subject 01, test on Subjects 02–05
Metric	Pearson correlation
Uncertainty	Bootstrap 95% confidence intervals
Keep ratios	0.10, 0.25, 0.50, 0.75, 1.00

Feature Pruning Strategies

We compare full feature representations with several feature selection strategies to identify optimal compact subsets.

01

Full Features

Uses all extracted features without any pruning or selection.

Baseline representation

02

Random Pruning

Keeps a completely random subset of features to serve as a strict control baseline.

03

Global Variance

Keeps features with high global variance. Implements a generic pruning baseline.

04

PROPOSED

ROI-Aware

Selects features based on ROI-specific Ridge coefficients. Our main proposed method for extreme compactness.

05

PROPOSED

Semantics-Aware

Uses semantic image groups to guide feature selection and interpret localized ROI responses.

Model Evaluation

Main metric: Pearson correlation between predicted and true ROI-level fMRI responses.

Higher Pearson r : Indicates superior alignment between model-driven predictions and empirically measured BOLD fMRI responses.

Baseline Comparison: Overall prediction accuracy is systematically benchmarked against the unpruned, full-feature model.

Retained Performance: Quantifies the precise percentage of baseline visual encoding capabilities preserved after aggressive feature pruning.

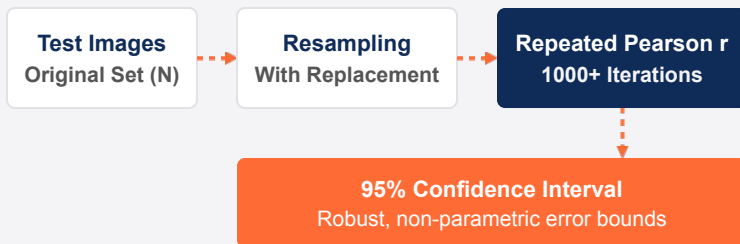
Bootstrap Confidence Intervals: Rigorously estimates prediction uncertainty by non-parametrically resampling the set of test stimuli.

Retained Performance Metric

$100 \times \text{Pruned } r / \text{Full Model } r$

Measures efficiency trade-off of compact feature subsets

Uncertainty Estimation via Bootstrapping



Main Results Summary

Quantitative comparison of feature pruning strategies across evaluation paradigms.

Evaluation	Full features mean r	ROI-aware 0.50 mean r	ROI-aware 0.25 mean r	ROI-aware 0.50 retained (%)	ROI-aware 0.25 retained (%)	Best Semantic Configuration	Best semantic mean r	Best semantic retained (%)
Cross-subject	0.4275	0.4273	0.4210	99.95%	98.49%	Semantics-aware pruning keep=0.75 thr=0.35	0.4303	100.65%
Within-subject	0.5098	0.5018	0.4756	98.43%	93.29%	ROI + semantics-aware pruning keep=0.75 thr=0.30	0.5076	99.56%

Cross-subject = train subj01, test subj02–05; Within-subject = train/test subj01.



Key Takeaway

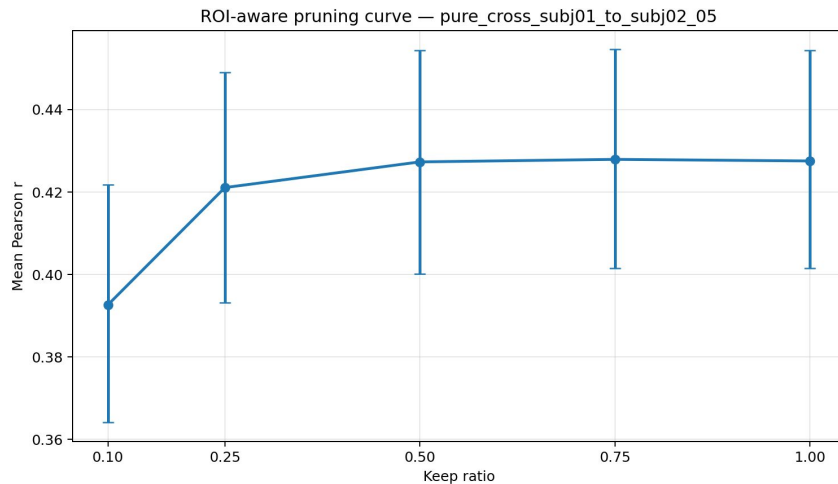
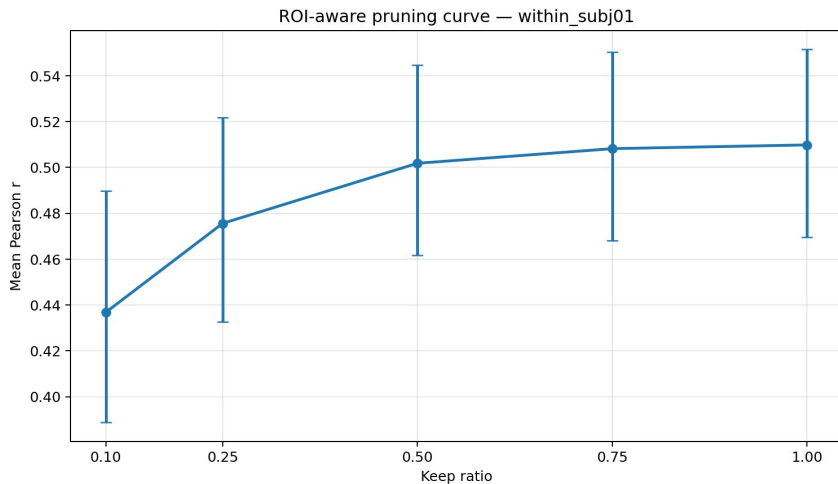
ROI-aware pruning at 50% keep ratio preserves almost all full-feature performance: 99.95% cross-subject and 98.43% within-subject.



Cautionary Note

The main contribution is performance preservation with fewer features, not a clear improvement over the full model. The semantic configurations are useful mainly for interpretability and show only small quantitative differences.

ROI-aware Pruning Curves



Key observations:

- Performance increases systematically with the feature keep ratio.
- At a **50% keep ratio**, ROI-aware pruning is already extremely close to full-feature baseline performance.
- Cross-subject generalization performance is especially stable and robust at the **25–50% keep ratio** range.

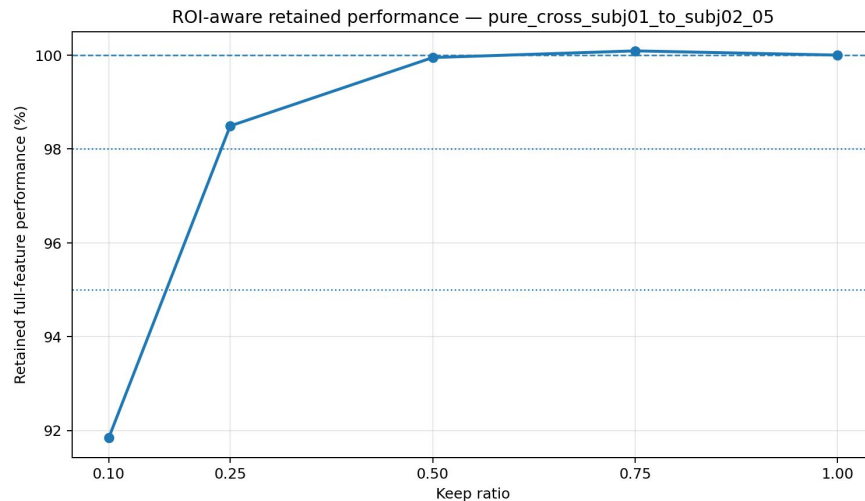
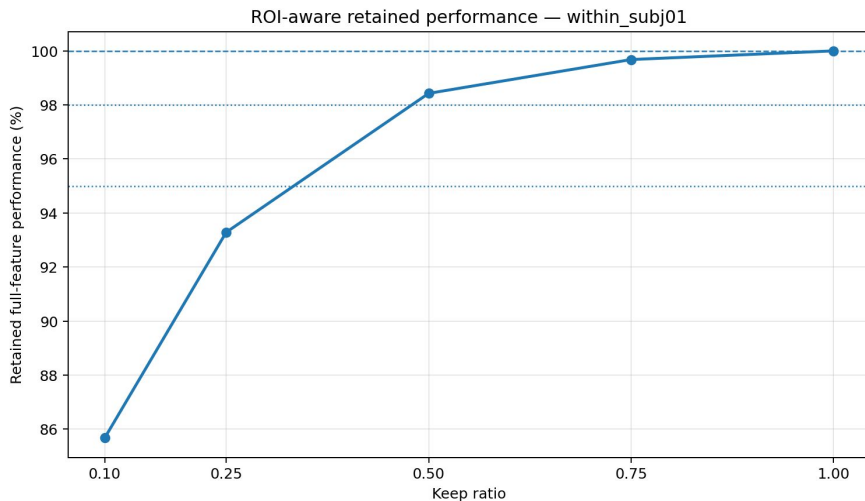


Final Takeaway

"Half of the features are enough to preserve almost the full ROI-level encoding performance."

Retained Performance vs Full Features

Comparing performance retention relative to the unpruned baseline across different keep ratios.



Metric Explanation:

Retained performance measures how much of the full-feature baseline is preserved after pruning. It directly quantifies the efficiency trade-off of compact feature subsets.

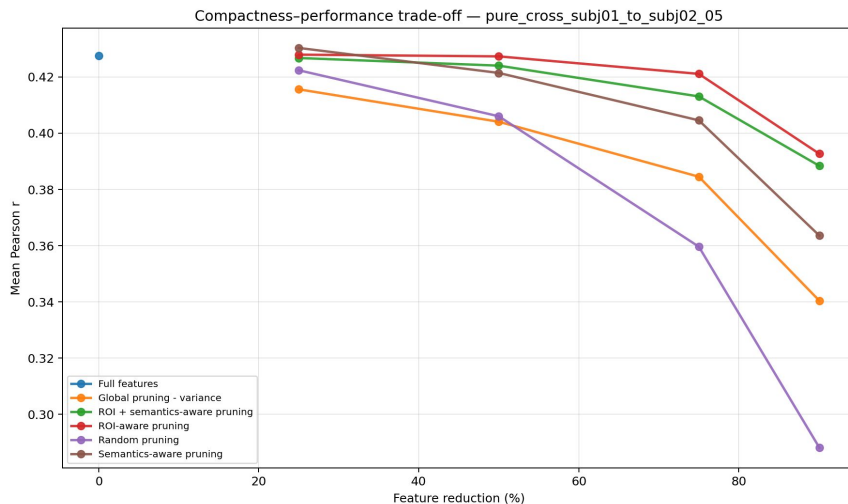
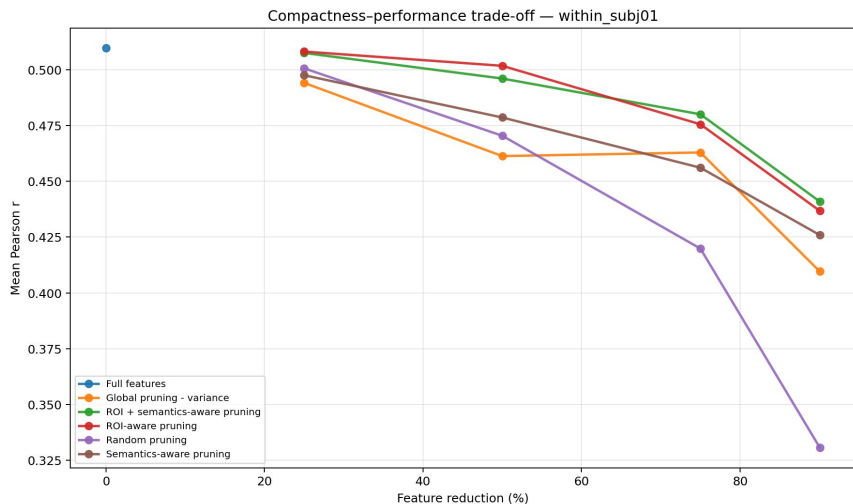
Key Finding



ROI-aware 50% retains 98.43% within-subject and 99.95% cross-subject performance.

Compactness-Performance Trade-off

Analyzing the balance between aggressive feature reduction and visual encoding performance retention.



Interpretation

"The contribution is not a higher absolute Pearson r, but a better compactness-performance trade-off."

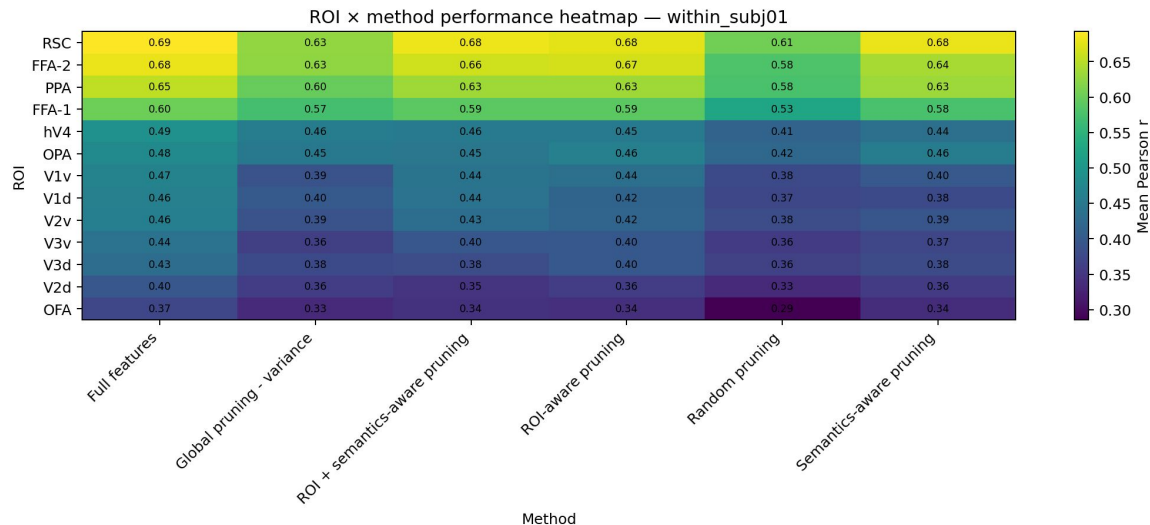
ROI-aware pruning successfully minimizes feature dimensionality while ensuring minimal performance degradation in the target brain regions.

ROI-aware 50% Key Metrics

- **50% feature reduction** achieved successfully.
- **98.43%** retained performance **within-subject**.
- **99.95%** retained performance **cross-subject**.

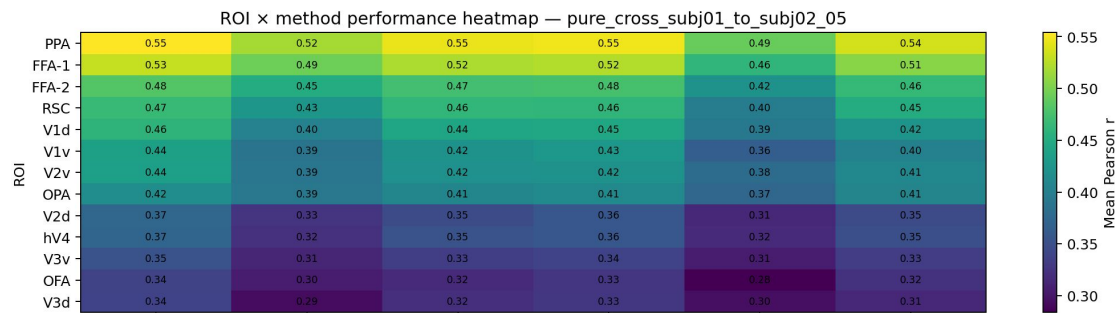
ROI-level Analysis

Detailed examination of feature pruning strategies across individual functional regions of interest.



Interpretation

- Different ROIs show **different prediction difficulty**, indicating varying baselines across visual regions.
- ROI-aware pruning is motivated by the fact that each functional ROI may rely on **distinct visual feature subsets**.
- The performance degradation or preservation effect of pruning is **not uniform** across cortical regions.

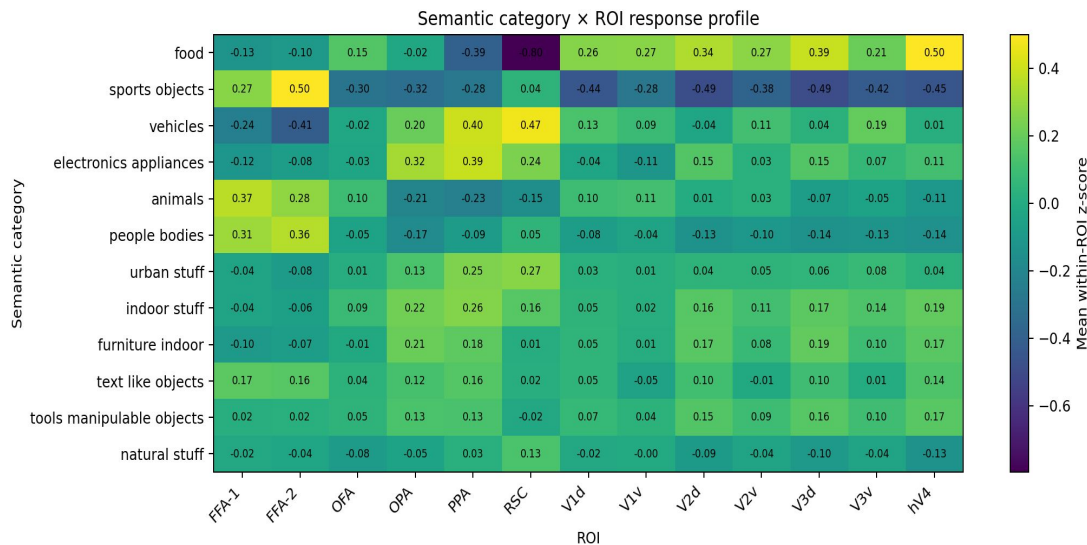


Key Takeaway

“ROI-specific feature selection is justified because visual ROIs do not behave identically.”

Semantic-aware ROI Response Analysis

Exploring localized cortical preferences and visual category processing across key brain regions.



Semantic Group	Best ROI	Interpretation
animals / bodies	FFA-1 / 2	high-level visual category preference
vehicles / urban	RSC	scene or navigation-related preference
indoor / furniture	PPA / OPA	place-related preference
food / tools	hV4	object-related visual response

Conclusions and Future Work

Synthesizing findings on compact, brain-aligned visual encoding and mapping next steps.

Conclusions

1. Pretrained visual features can effectively predict ROI-level fMRI responses.
2. ROI-aware pruning preserves almost all full-feature performance with fewer features.
3. At a 50% keep ratio, ROI-aware pruning retains about 98–100% of full-model performance.
4. Semantics-aware analysis is primarily useful for interpretability.
5. The project successfully supports compact and interpretable brain-aligned visual encoding.

Future Work

- Transition from ROI-level to voxel-level prediction
- Compare additional backbones and layers (DINOv2, CLIP)
- Integrate subject-aware adaptation and estimate noise ceiling
- Incorporate rigorous statistical significance tests



Key Takeaway

ROI-aware pruning does not clearly outperform full features, but it can remove a large fraction of features while preserving almost the same brain-encoding performance.

Key References

- VISGate: ROI-Conditioned Dual-Head Encoders
- BOLDREAMS: Dreaming with pruned in-silico fMRI models
- DINOv2 & CLIP Backbones
- Natural Scenes Dataset / Algonauts-style brain data